Response to Reviewers

We thank both reviewers for their careful reading of the manuscript. Below we address each comment in turn, with the corresponding change in the manuscript cited by page number.

Reviewer 1

Reviewer 1 — comment 1
Please specify which propensity score methods were actually used — "propensity score methods" alone is too vague.

We have specified that we used inverse probability weighting and matching. (modified on p. 1)

Reviewer 1 — comment 11
A single comment (illustrative, for testing) that required a change in two separate sections — the introduction and the conclusion.

Addressed in both locations: (modified on p. 1 and p. 3).

Shown as two separate excerpts, one per location:

p. 1 — This revision responds to a single comment that required changes in two separate sections of the manuscript — this sentence, in the introduction, is one of them.

p. 3 — This sentence, in the conclusion, was the second of the two changes made in response to that same comment.

Reviewer 1 — comment 2
How exactly were confounders selected? "Prior clinical knowledge" is not a reproducible criterion.

We now describe the disjunctive cause criterion, cited in the manuscript, as our primary selection method, with clinical knowledge used only as a secondary check.

p. 1 — ~~Confounders were selected based on prior clinical knowledge.~~ Confounders were selected using the disjunctive cause criterion described by [1], and prior clinical knowledge was used only as a secondary check.

Reviewer 1 — comment 3
Which target trial emulation framework was followed?

We now cite the cloning, censoring, and weighting approach, referenced in the manuscript, explicitly. (modified on p. 1)

Reviewer 1 — comment 4
Please clarify the censoring events used to define follow-up time.

We have kept the original wording here, which we believe already answers this point; see the highlighted passage.

p. 1 — Patients were followed from cohort entry until the earliest of death, loss to follow-up, or administrative censoring.

Reviewer 1 — comment 5
The propensity score model as written does not condition on time of entry, which matters for a target trial emulation with staggered entry.

We agree and have corrected the propensity score definition to condition explicitly on t_0, and we now also use restricted cubic splines for continuous covariates.

p. 1 — The propensity score was estimated as

$$\pi(x) = P(A = 1 \mid X = x, t_0) \tag{1}$$

using logistic regression with restricted cubic splines for continuous covariates.

Reviewer 1 — comment 6
Table 1 should report the standardized mean difference, not just raw means, to demonstrate balance.

We have adjusted the "after weighting" age estimate for consistency with the balance figure; a full standardized mean difference table is shown in Fig. 1.

p. 2 —

Covariate	Before weighting	After weighting
Age, years	54.2 (12.1)	54.1 (12.0)
Sex, % female	48.3	48.1
Diabetes, %	22.7	21.9

Table I: Baseline characteristics before and after weighting.

Reviewer 1 — comment 9
The calibration equation seems redundant with equation 1.

We agree and have removed it.

Removed: l'équation de calibration, devenue redondante avec l'équation 1.

Reviewer 1 — comment 10
Please report sensitivity to the truncation threshold.

This was explored early on but is redundant with the E-value analysis and has been removed for concision.

Removed: le tableau de sensibilité au seuil de troncature.

Reviewer 1 — comment 7
An unmeasured confounding sensitivity analysis is missing.

An E-value sensitivity analysis has been added, following the recommendations cited in the manuscript. (modified on p. 2)

Reviewer 1 — comment 8
The conclusion should not overstate causal language given the observational design.

We have reviewed the conclusion and believe the current wording, already qualified throughout the discussion, remains appropriate; we have therefore left it unchanged.

p. 3 — We conclude that early treatment initiation was associated with improved outcomes in this emulated trial.

Reviewer 2

Reviewer 2 — comment 1
The methods section heading should indicate that methods were updated relative to the registered protocol.

The heading has been updated accordingly.

p. 1 — Méthodes actualisées

Reviewer 2 — comment 2
Please add a covariate balance figure (e.g. a Love plot) to support the claim that weighting achieved adequate balance.

A standardized mean difference figure has been added.

p. 2 —

Figure 1: Standardized mean differences before and after weighting.

Reviewer 2 — comment 3
The per-protocol analysis is not pre-specified and, given space constraints, could be removed to focus the paper on the primary intention-to-treat-style emulation.

We agree and have removed the per-protocol subsection in its entirety, including its figure and table.

Removed: l'ancienne analyse per-protocole, y compris la figure et le tableau associés.

Reviewer 2 — comment 4
The discussion's caution about the observational design could acknowledge the target trial framework's mitigating role more explicitly.

We have expanded this sentence accordingly, as already discussed in our response to reviewer 1, comment 2, p. 1 on the choice of confounder selection criterion.

p. 3 — These results should be interpreted in light of the observational design, though the target trial framework mitigates several sources of bias common to such analyses.

Editor

⚠ comment e1 has no matching revision in the manuscript **Editor — comment 1**

Please ensure all figures referenced in the text include a page citation in the response letter, per journal policy.

Done throughout; see e.g. the balance figure, referenced at Fig. 1, p. 2, and the baseline table, referenced at Table I, p. 2.

For the reviewers only

The figure below summarizes, for the reviewers' convenience only, how many comments from each reviewer led to a textual change versus a clarification without a change to the manuscript.

Reviewer	Comments	Textual changes
Reviewer 1	8	6
Reviewer 2	4	4

Table R2: Summary of comments and changes, for the reviewers' convenience.

As shown in Table R2, most comments led to a direct textual change, with a smaller number addressed by clarification alone (as discussed by S. Schneeweiss [2] in a related methodological context). We also note the sensitivity-analysis literature more broadly A. Jones and W. Ng [3].

References cited in this response

[1] T. J. VanderWeele, "Principles of Confounder Selection," *European Journal of Epidemiology*, 2019.

[2] S. Schneeweiss, "Automated Data-Adaptive Analytics," *Epidemiology*, 2018.

[3] A. Jones and W. Ng, "On Sensitivity Analyses for Unmeasured Confounding," *Journal of Causal Methods*, 2021.